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Studierichting: Informatica
Oefeningenreeks 2D nr. 17.2

$$f(x) = \frac{1 + \cos 2x}{\sin x}$$
$$= \frac{1 + \left(1 - 2\sin^2 x\right)}{\sin x}$$
$$= \frac{2 - 2\sin^2 x}{\sin x}$$
$$= 2\frac{1 - \sin^2 x}{\sin x}$$
$$= 2\frac{\cos^2 x}{\sin x}$$
$$\lim_{x \to \frac{\pi}{2}} f(x)$$
$$= 2\frac{\cos^2\left(\frac{\pi}{2}\right)}{\sin\left(\frac{\pi}{2}\right)}$$
$$= 2\frac{0}{1}$$
$$= 0$$

Voor $x \to 0 : \lim_{x \to 0}\left(\frac{1 + \cos(2x)}{\sin(x)}\right) = \frac{1 + \cos(0)}{\sin(0)} = \frac{1+1}{0} = \infty$

$$\lim_{x \to 0+}\left(\frac{1 + \cos(2x)}{\sin(x)}\right) = \frac{1 + \cos(0+)}{\sin(0+)} = \frac{1 + 1+}{0+} = +\infty$$
$$\lim_{x \to 0-}\left(\frac{1 + \cos(2x)}{\sin(x)}\right) = \frac{1 + \cos(0-)}{\sin(0-)} = \frac{1 + 1+}{0-} = -\infty$$

Voor $x \to \pi : \lim_{x \to \pi}\left(\frac{1 + \cos(2x)}{\sin(x)}\right) = \frac{1 + \cos(2\pi)}{\sin(\pi)} = \frac{1+1}{0} = \infty$

$$\lim_{x \to \pi+}\left(\frac{1 + \cos(2x)}{\sin(x)}\right) = \frac{1 + \cos(2\pi+)}{\sin(\pi+)} = \frac{1 + 1+}{0-} = -\infty$$
$$\lim_{x \to \pi-}\left(\frac{1 + \cos(2x)}{\sin(x)}\right) = \frac{1 + \cos(2\pi-)}{\sin(\pi-)} = \frac{1 + 1+}{0+} = +\infty$$

References